Akashic Civics

From Invisible Injustice to Measurable Democracy

Akashic Civics Project Team

The Civic Problem: Democracy Without Data

Local criminal justice decisions shape lives – but they are invisible. District attorneys, sheriffs, and judges exercise enormous discretion over arrests, charges, bonds, sentences, and parole. Race, income, and neighborhood predict outcomes more reliably than the law should. Yet without measurement, communities cannot name it, cannot challenge it, cannot change it.
Why justice requires measurement: Transparency is the minimum condition for accountability in a democracy.

What We're Building: Tools for Community Power

Akashic Civics is open-source infrastructure that helps communities turn fragmented local data into actionable insight. We combine statistical rigor with civic accessibility: `TypeScript` → `Python` → `R`, modular extensions, and dashboards designed for journalists, researchers, and activists.
The vision: Data belongs to the community, not to the prosecutor.

Design Principles: Multi-language, Statistical, Composable

Two principles drive everything we build:
(1) **Multi-language by design** – Meet users where they are (policy researchers use R, data scientists use Python, civic tech uses TypeScript).
(2) **Statistical rigor first** – No headline numbers without confidence intervals and causal interpretation.
(3) **Composable, not monolithic** – Use the dashboard, the CLI, or just the R package.
Why this matters: Tools must fit into existing practices, not reinvent them.

E16: Data Integration – The Foundation of Everything

Data hides in silos. FBI UCR, BJS, and local court systems store 90% of what we need – but in incompatible formats. Pluggable `DataSource` interfaces and a `DataPipeline` that normalizes to a common schema make analysis possible.
For democracy: If data stays trapped in silos, communities stay powerless. 9 tests ensure reliability across data sources.

E17: Advanced Analytics – Causal, Not Just Correlational

Rates lie. A 1.4× conviction rate disparity might mean bias – or just different crimes. `DoublyRobustEstimator` protects against model misspecification; `BayesianHierarchicalModel` stabilizes estimates in small jurisdictions.
For justice: Causal methods answer "Did this policy cause disparity?" not "Are they correlated?" This is how we move from observation to evidence-based reform.

E18: Interactive CLI Wizard – Tools for Everyone

No PhD required. Non-programmers can load data, choose Jurisdiction, select analysis type, and get results in five steps. The wizard abstracts technical complexity while preserving methodological choices.
For democracy: When tools are usable, power diffuses. A civic organizer, not a statistician, can

make the case to their DA.

E19: Jupyter Notebook Interface – The Analyst's Playground

Python users want magic, not CLI flags. IPython magics (`%akashic`), rendered report templates, and direct NumPy/SciPy integration let analysts iterate interactively.
For justice: Analysts can prototype, explore, and build public-facing narratives. When research is reproducible in a notebook, findings become shareable across coalitions.

E20: Publishing Pipeline – Open Source Cannot Collapse

Open-source projects die. An 11-step release pipeline (`npm publish`, `pip install`, `R CMD INSTALL`, `docker push`, automated testing, changelogs) keeps code current and trustworthy.
For democracy: Sustainability ensures tools survive grant cycles and volunteer turnover. Code must be a public good, not a one-person effort.

E21: Extension System – Communities Add Their Own Power

Needs differ. A Native American tribal court needs different data sources than urban DA's office. `ExtensionRegistry` lets communities ship custom plugins for DataSources, Analyses, and Exports.
For democracy: Tools adapt to local contexts instead of requiring local coding skills. Community ownership starts with modular architecture.

E22: R Package – Speaking Policy's Language

Policy researchers think in models, not APIs. The `akashic.civics` R package provides OLS, logistic regression, doubly-robust estimation, and disconfirmation batteries familiar to criminologists and political scientists. `R CMD check` compliant, with vignettes.
For justice: When academic researchers can run the same analysis, findings carry weight in scholarly and policy debates.

E23: Streaming Analysis – Justice in (Near) Real Time

Disparity moves fast. Policy changes, court closures, or surges in specific crime types can shift ratios dramatically. `StreamProcessor` reads JSONL in chunks, tracks group-level statistics, and fires alerts when disparity crosses thresholds.
For democracy: Real-time monitoring turns reactive analysis into proactive intervention. When the ratio jumps, advocates know to act.

E24: Web Dashboard UI – Storytelling for Citizens

Most users won't run code. An Express REST API (`/api/health`, `/api/viz-specs`, `/api/analyze`) powers a static single-page app showing disparity ratios, conviction rates, and sentence lengths.
For justice: When disparities are visible on a dashboard in a community meeting, officials must confront them.

E25: Python Statistical Depth – Scientific Rigor, Not Guesswork

Numbers must be right. Real NumPy/SciPy implementations of descriptive statistics, Z-tests, OLS, logistic regression, doubly-robust estimation, and disconfirmation batteries replace placeholder functions. Error handling with informative messages.
For democracy: Communities can trust that their analysis withstands peer review, Freedom of Information Act challenges, and expert scrutiny.

A Concrete Democracy in Action

Scenario. Mid-sized county, 12,000 cases/year. A community group loads 3 years of data into Akashic Civics. Doubly-robust analysis: Black defendants convicted at 1.4× rate of White defendants ($p < 0.01$). Streaming alert: ratio jumped from 1.2× to 1.6× after new DA took office. Group presents findings at public safety committee hearing. DA adopts new charging guidelines. Next quarter: disparity drops to 1.1×.

Key: Measurement enabled accountability; accountability drove policy change; policy change reduced disparity.

Toward a More Just Democracy

- **Transparency:** Disparity metrics no longer locked in prosecutors' spreadsheets
- **Participation:** Community groups analyze the same data as officials
- **Accountability:** Elected leaders evaluated on measurable outcomes
- **Reform Evaluation:** Did the policy actually reduce disparity? We can tell.

Get Involved

- Install: `npm install akashic-civics` | `pip install akashic-civics-py` | `R CMD INSTALL`
- Run: `akashic wizard`, `akashic dashboard`
- Contribute: extensions, data sources, visualizations, documentation
- GitHub: `github.com/elijah/akashic-civics`

Questions & Discussion

> What local decisions most need measurement in your community? Which data sources would you ship as extensions? How can tools move from reporting injustice to changing it?